Overall Model Performance

In [4]:
```
import pandas as pd
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, c

file_path = r"C:\Users\terbe\T_RS_STUDENT_2023.parquet"
df = pd.read_parquet(file_path)

# Filter DataFrame
df_fall = df[df['TERM_CD'] == '220238']
df_fall = df_fall[df_fall['STUDENT_TYPE_CD'] == 'C']
# Drop duplicates
df_fall['EDW_PERS_ID'] = df_fall['EDW_PERS_ID'].drop_duplicates()

# Convert to string, remove ".0" suffix, and convert to int
df_fall['EDW_PERS_ID'] = df_fall['EDW_PERS_ID'].astype(str).str.replace('.0', '').asty

# Get lists
enrolled_continued_fall_list = list(set(df_fall['EDW_PERS_ID']))

original_snapshot = pd.read_parquet("COMBINED_TARGET_BY_YEAR.parquet")
original_snapshot_2223 = original_snapshot[
    original_snapshot['TERM_CD'].isin(['220231', '220235', '220228'])
]

# Convert 'DEG_STATUS_CD' to string and then filter
awarded_degree_students_2223_df = original_snapshot_2223[
    original_snapshot_2223['DEG_STATUS_CD'].astype(str).eq('AW')
]


students_to_graduate = list(set((list(awarded_degree_students_2223_df["EDW_PERS_ID"]))

not_conitinuing = df[~(df['STUDENT_TYPE_CD'] == 'C')]
not_conitinuing['STUDENT_TYPE_DESC'].value_counts()

# Convert to string, remove ".0" suffix, and convert to int
df['EDW_PERS_ID'] = df['EDW_PERS_ID'].astype(str).str.replace('.0', '').astype(int)
df_2023 = df[df['TERM_CD'].isin(['220231', '220235', '220228'])]
students_to_continue = list(set((list(df_2023["EDW_PERS_ID"]))))

# Assuming df_2023 is your DataFrame
df_2023['TERM_CD'] = df_2023['TERM_CD'].astype(int)

grouped_df = df_2023.groupby('EDW_PERS_ID')['TERM_CD'].idxmax()
df_2023_filtered = df_2023.loc[grouped_df]

sheet_names = ['Pivot_All_Discontinued', 'Pivot_Undergrads', 'Pivot_Grads', 'Pivot_Pro

# Read each sheet into a separate dataframe
ids_discontinued_dfs = pd.read_excel('EDW_PERS_ID_Students_to_Discontinue.xlsx', sheet

# Assign each dataframe to its respective variable
pivot_all_discontinued_df = ids_discontinued_dfs.get(sheet_names[0])
pivot_undergrads_df = ids_discontinued_dfs.get(sheet_names[1])
pivot_grads_df = ids_discontinued_dfs.get(sheet_names[2])
```

```
pivot_professional_df = ids_discontinued_dfs.get(sheet_names[3])
pivot_law_df = ids_discontinued_dfs.get(sheet_names[4])
all_discontinued_df = ids_discontinued_dfs.get(sheet_names[5])
undergrads_discontinued_df = ids_discontinued_dfs.get(sheet_names[6])
grads_discontinued_df = ids_discontinued_dfs.get(sheet_names[7])
professional_discontinued_df = ids_discontinued_dfs.get(sheet_names[8])
law_discontinued_df = ids_discontinued_dfs.get(sheet_names[9])

pred_discontinued_students = list(set(all_discontinued_df['EDW_PERS_ID']))
student_2223 = list(set(list(df_2023["EDW_PERS_ID"])))

df_predictions = pd.DataFrame()
df_predictions["EDW_PERS_ID"] = student_2223

# Remove rows where EDW_PERS_ID is in students_to_graduate
df_predictions = df_predictions[~df_predictions["EDW_PERS_ID"].isin(students_to_gradua

df_predictions['RF_PREDICTION_TO_CONTINUE'] = df_predictions["EDW_PERS_ID"].apply(lamb
df_predictions["ACTUAL_CONTINUED"] = df_predictions["EDW_PERS_ID"].apply(lambda x: 1 i

validation = []
is_correct = []

# Loop through each row in the DataFrame using iterrows
for index, row in df_predictions.iterrows():
    if row["RF_PREDICTION_TO_CONTINUE"] == 1 and row['ACTUAL_CONTINUED'] == 1:
        validation.append("true continued")
        is_correct.append(1)
    elif row["RF_PREDICTION_TO_CONTINUE"] == 0 and row['ACTUAL_CONTINUED'] == 0:
        validation.append("true discontinued")
        is_correct.append(1)
    elif row["RF_PREDICTION_TO_CONTINUE"] == 0 and row['ACTUAL_CONTINUED'] == 1:
        validation.append("false continued")
        is_correct.append(0)
    elif row["RF_PREDICTION_TO_CONTINUE"] == 1 and row['ACTUAL_CONTINUED'] == 0:
        validation.append("false discontinued")
        is_correct.append(0)

# Add the new lists to the DataFrame
df_predictions['Validation'] = validation
df_predictions['Is_Correct'] = is_correct

# Extract the actual and predicted values
actual_values = df_predictions['ACTUAL_CONTINUED']
predicted_values = df_predictions['RF_PREDICTION_TO_CONTINUE']

# Calculate confusion matrix
conf_matrix = confusion_matrix(actual_values, predicted_values)

# Extract True Positives, True Negatives, False Positives, and False Negatives
tn, fp, fn, tp = conf_matrix.ravel()

# Calculate Accuracy
accuracy = accuracy_score(actual_values, predicted_values)

# Calculate Precision
precision = precision_score(actual_values, predicted_values)

# Calculate Recall (Sensitivity)
recall = recall_score(actual_values, predicted_values)
```

```python
# Calculate F1 Score
f1 = f1_score(actual_values, predicted_values)


# Create a DataFrame
results_df = pd.DataFrame({
    'Metric': ['Confusion Matrix', 'Accuracy', 'Precision', 'Recall', 'F1 Score'],
    'Value': [conf_matrix, accuracy, precision, recall, f1]
})

# Display the DataFrame
results_df
```

```
C:\Users\terbe\AppData\Local\Temp\ipykernel_30280\949066921.py:41: SettingWithCopyWar
ning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/us
er_guide/indexing.html#returning-a-view-versus-a-copy
  df_2023['TERM_CD'] = df_2023['TERM_CD'].astype(int)
```

Out[4]:

	Metric	Value
0	Confusion Matrix	[[1840, 2640], [1204, 19254]]
1	Accuracy	0.845858
2	Precision	0.879419
3	Recall	0.941148
4	F1 Score	0.909237

Assessment

Confusion Matrix:

```
True Positives (TP): 19254
True Negatives (TN): 1850
False Positives (FP): 2640
False Negatives (FN): 1204
```

Accuracy:

```
Accuracy: 0.845858
```

Accuracy is the proportion of correctly classified instances. An accuracy of approximately 84.3% suggests that the model is correct in its predictions for around 84.3% of the instances. In binary classification tasks, a good accuracy value depends on the specific context, but generally, a value above 80% is considered decent. High accuracy alone may not be sufficient if the dataset is imbalanced.

Recall (Sensitivity):

```
Recall: 0.941148
```

Recall (Sensitivity) is the proportion of true positive predictions among all actual positives. A recall of approximately 93.1% suggests that the model is able to correctly identify about 93.1% of the instances with the positive outcome. In the context of predicting student continuance, a high recall is important as it indicates a low rate of false negatives – instances where the model fails to identify students who will continue. A good recall value depends on the trade-off with precision, but generally, values above 90% are considered favorable.

Precision:

```
Precision: 0.879419
```

Precision is the proportion of true positive predictions among all instances predicted as positive. A precision of approximately 87.9% means that when the model predicts a positive outcome, it is correct about 87.9% of the time. In the context of predicting student continuance, a high precision is desirable as it indicates a low rate of false positives – instances where the model incorrectly predicts a student will continue when they do not. A good precision value depends on the specific goals of the model, but values above 80% are often considered satisfactory.

F1 Score:

```
F1 Score: 0.909237
```

F1 Score is the harmonic mean of precision and recall. An F1 Score of approximately 90.4% indicates a balance between precision and recall. It is a useful metric when both false positives and false negatives are important. A good F1 Score depends on the specific requirements of the task, but values above 80% are generally considered good. In the context of predicting student continuance, a high F1 Score suggests a well-balanced model performance.

Assessment:

The high accuracy, precision, recall, and F1 Score suggest that the model is performing well overall. The model has a good balance between precision and recall, as indicated by the relatively high F1 Score. The high recall suggests that the model is effective in identifying positive instances (those who actually continued). The precision indicates that when the model predicts a positive outcome, it is correct the majority of the time. In summary, based on these metrics, the model appears to be of good quality for the given binary classification task.

In [11]:
```
import pandas as pd
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_

student_level_list = list(df_updated["STUDENT_LEVEL_DESC"].unique())
coll_name_list = list(df_updated["ACAD_COLL_NAME"].unique())

# Function to calculate metrics
def calculate_metrics(df, label_name):
    actual_values = df['ACTUAL_CONTINUED']
    predicted_values = df['RF_PREDICTION_TO_CONTINUE']
    conf_matrix = confusion_matrix(actual_values, predicted_values)
    tn, fp, fn, tp = conf_matrix.ravel()
    accuracy = accuracy_score(actual_values, predicted_values)
    precision = precision_score(actual_values, predicted_values)
    recall = recall_score(actual_values, predicted_values)
    f1 = f1_score(actual_values, predicted_values)

    results_df = pd.DataFrame({
        'Confusion_Matrix': [conf_matrix],
        'Accuracy': [accuracy],
        'Precision': [precision],
        'Recall': [recall],
        'F1 Score': [f1],
        'True Positive': [tp],
        'False Positive': [fp],
        'True Negative': [tn],
        'False Negative': [fn],
        'Label Name': [label_name]
    })

    return results_df

# Drop duplicates from df_predictions based on 'EDW_PERS_ID'
df_predictions_unique = df_predictions.drop_duplicates(subset='EDW_PERS_ID')

# Left join and select specific columns from df_fall
merged_df = df_predictions_unique.merge(df_2023_filtered[['EDW_PERS_ID', 'STUDENT_LEVE
                                        on='EDW_PERS_ID', how='left')

df_updated = merged_df.copy()

# Define columns for grouping
grouping_columns = ["STUDENT_LEVEL_DESC", "ACAD_COLL_NAME"]

# Initialize an empty list to store results DataFrames
result_dataframes = []

# Loop through each grouping column
for col in grouping_columns:
    # Get unique values for the current grouping column
    unique_values = df_updated[col].unique()

    # Loop through each unique value in the grouping column
    for val in unique_values:
```

```
        # Filter the DataFrame based on the current unique value
        filtered_df = df_updated[df_updated[col] == val]

        # Calculate metrics using the function
        result_df = calculate_metrics(filtered_df, f"{col}_{val}")

        # Append the result DataFrame to the list
        result_dataframes.append(result_df)

# Concatenate all DataFrames along rows (axis=0)
result = pd.concat(result_dataframes, axis=0, ignore_index=True)

# Sort the DataFrame by 'Accuracy' column in descending order
result = result.sort_values(by='Accuracy', ascending=False).reset_index(drop=True)

# Display the result DataFrame
result
```

Out[11]:

	Confusion_Matrix	Accuracy	Precision	Recall	F1 Score	True Positive	False Positive	True Negative	False Negative
0	[[0, 1], [0, 72]]	0.986301	0.986301	1.000000	0.993103	72	1	0	0
1	[[2, 17], [6, 500]]	0.956190	0.967118	0.988142	0.977517	500	17	2	6
2	[[0, 12], [2, 279]]	0.952218	0.958763	0.992883	0.975524	279	12	0	2
3	[[28, 93], [33, 1846]]	0.937000	0.952037	0.982437	0.966998	1846	93	28	33
4	[[9, 10], [3, 173]]	0.933333	0.945355	0.982955	0.963788	173	10	9	3
5	[[7, 39], [2, 497]]	0.924771	0.927239	0.995992	0.960386	497	39	7	2
6	[[1, 16], [2, 135]]	0.883117	0.894040	0.985401	0.937500	135	16	1	2
7	[[229, 385], [151, 3610]]	0.877486	0.903630	0.959851	0.930892	3610	385	229	151
8	[[197, 29], [1, 0]]	0.867841	0.000000	0.000000	0.000000	0	29	197	1
9	[[62, 104], [35, 848]]	0.867493	0.890756	0.960362	0.924251	848	104	62	35
10	[[1, 20], [7, 174]]	0.866337	0.896907	0.961326	0.928000	174	20	1	7
11	[[27, 42], [52, 547]]	0.859281	0.928693	0.913189	0.920875	547	42	27	52
12	[[27, 42], [52, 547]]	0.859281	0.928693	0.913189	0.920875	547	42	27	52
13	[[1064, 1602], [931, 13358]]	0.850605	0.892914	0.934845	0.913399	13358	1602	1064	931
14	[[195, 397], [188, 2953]]	0.843290	0.881493	0.940146	0.909875	2953	397	195	188
15	[[652, 769], [503, 6157]]	0.842594	0.888969	0.924474	0.906374	6157	769	652	503
16	[[50, 89], [25, 488]]	0.825153	0.845754	0.951267	0.895413	488	89	50	25
17	[[140, 628], [110, 3272]]	0.822169	0.838974	0.967475	0.898654	3272	628	140	110
18	[[126, 206], [71, 1073]]	0.812331	0.838937	0.937937	0.885679	1073	206	126	71
19	[[50, 124], [41, 651]]	0.809469	0.840000	0.940751	0.887526	651	124	50	41
20	[[39, 6], [13, 32]]	0.788889	0.842105	0.711111	0.771084	32	6	39	13
21	[[18, 59], [10, 234]]	0.785047	0.798635	0.959016	0.871508	234	59	18	10
22	[[58, 27], [8, 63]]	0.775641	0.700000	0.887324	0.782609	63	27	58	8

	Confusion_Matrix	Accuracy	Precision	Recall	F1 Score	True Positive	False Positive	True Negative	False Negative
23	[[109, 166], [82, 720]]	0.769731	0.812641	0.897756	0.853081	720	166	109	82
24	[[230, 47], [52, 91]]	0.764286	0.659420	0.636364	0.647687	91	47	230	52
25	[[347, 85], [24, 6]]	0.764069	0.065934	0.200000	0.099174	6	85	347	24
26	[[4, 51], [2, 130]]	0.716578	0.718232	0.984848	0.830671	130	51	4	2
27	[[8, 30], [2, 44]]	0.619048	0.594595	0.956522	0.733333	44	30	8	2
28	[[0, 92], [0, 4]]	0.041667	0.041667	1.000000	0.080000	4	92	0	0
29	[[0, 92], [0, 4]]	0.041667	0.041667	1.000000	0.080000	4	92	0	0

In [12]:

```
# Assuming 'Evaluation_Metrics_Fall_2023.xlsx' is the desired file name
result.to_excel('Evaluation_Metrics_Fall_2023.xlsx', index=False)
```

Assessment

High Accuracy and Precision:

Coll of Med Office of the Dean:

Exceptional accuracy and precision (both 98.63%) highlight the model's robust performance in predicting student continuation for medical office-related programs.

Coll Medicine at Chicago - CS:

Demonstrates high accuracy (95.62%) and precision (96.71%), signifying reliable predictions for medicine-related programs at Chicago - CS.

Dentistry:

Strong accuracy (95.22%) and precision (95.88%) showcase the model's effectiveness in predicting student continuity for Dentistry programs

Challenges and Opportunities for Improvement:

VP Undrgrd Affrs & Acdmc Prgms:

Challenges are evident with 86.78% accuracy, where precision is notably absent. Given the small population size, it is reasonable to attribute the lower accuracy to this factor, and further investigation may not be necessary.

Undergrad Non-Degree Chicago:

Presents challenges with 76.41% accuracy, lower precision (6.59%), and recall (20.00%). The small population size is likely the main contributing factor to the lower accuracy, and this subset might not require extensive investigation beyond acknowledging this limitation.

Contextual Considerations:

Non-Credit - Chicago, VP for Global Engagement:

Extreme cases with minimal samples and 4.17% accuracy emphasize the importance of considering the context and potential biases for certain subsets.

Summary

In conclusion, the model exhibits exceptional performance in subsets related to the medical and professional domains, reflected in high accuracy and precision. However, challenges are evident in subsets characterized by smaller populations, such as "VP Undrgrd Affrs & Acdmc Prgms" and "Undergrad Non-Degree Chicago," where accuracy may be impacted by the inherent difficulties of predicting with limited data.

Contextual understanding is crucial in the interpretation of results, considering specific goals and constraints. It's important to acknowledge that subsets with lower population sizes may naturally exhibit lower accuracy. Continued monitoring and iterative refinement of the model are recommended to enhance overall predictive capabilities. Tailoring strategies based on distinct predictability patterns observed across different academic colleges can further optimize the model's performance. This holistic approach ensures a nuanced evaluation and ongoing improvement of the machine learning model for predicting student continuation into the fall semester.

In [13]:
```
import pandas as pd
import numpy as np
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_

# Assuming df_updated, student_level_list, and other required variables are defined
def calculate_metrics(df, label_name):
    actual_values = df['ACTUAL_CONTINUED']
    predicted_values = df['RF_PREDICTION_TO_CONTINUE']

    # Calculate confusion matrix
    conf_matrix = confusion_matrix(actual_values, predicted_values)

    if conf_matrix.shape == (1, 1):  # Binary classification with only one class
        # Create a 2x2 matrix with appropriate values
        conf_matrix = np.array([[0, 0], [0, conf_matrix[0, 0]]])

    # Extract True Positives, True Negatives, False Positives, and False Negatives
    tn, fp, fn, tp = conf_matrix.ravel()

    # Calculate Accuracy
    accuracy = accuracy_score(actual_values, predicted_values)
```

```
    # Calculate Precision
    precision = precision_score(actual_values, predicted_values)

    # Calculate Recall (Sensitivity)
    recall = recall_score(actual_values, predicted_values)

    # Calculate F1 Score
    f1 = f1_score(actual_values, predicted_values)

    # Create a DataFrame
    results_df = pd.DataFrame({
        'Confusion_Matrix': [conf_matrix],
        'Accuracy': [accuracy],
        'Precision': [precision],
        'Recall': [recall],
        'F1 Score': [f1],
        'True Positive': [tp],
        'False Positive': [fp],
        'True Negative': [tn],
        'False Negative': [fn],
        'Label Name': [label_name]
    })

    return results_df


# Create an empty dictionary to store DataFrames
dataframes_dict = {}

# Iterate over each student level
for student_level in student_level_list:
    # Filter the DataFrame based on student level
    undergraduate_df = df_updated[df_updated['STUDENT_LEVEL_DESC'] == student_level]

    # Get unique academic collection names for the current student level
    undergraduate_coll_name_list = undergraduate_df["ACAD_COLL_NAME"].unique()

    # Create DataFrames for each academic collection name and store them in the dictic
    for coll_name in undergraduate_coll_name_list:
        current_df = undergraduate_df[undergraduate_df["ACAD_COLL_NAME"] == coll_name]

        # Create a key for the dictionary using a formatted string
        key = f"{student_level}_{coll_name}"

        # Store the DataFrame in the dictionary
        dataframes_dict[key] = current_df

# Now you can access each DataFrame using the dictionary keys
# For example, dataframes_dict["Undergraduate_Liberal Arts & Sciences"] will give you

# Calculate metrics for each DataFrame and store results in another dictionary
metrics_dict = {}
for key, df in dataframes_dict.items():
    metrics_df = calculate_metrics(df, key)
    metrics_dict[key] = metrics_df

# Concatenate all metrics DataFrames along rows (axis=0)
result = pd.concat(metrics_dict.values(), axis=0, ignore_index=True)
```

```
# Sort the DataFrame by 'Accuracy' column in descending order
coll_student_level_result = result.sort_values(by='Accuracy', ascending=False).reset_i
coll_student_level_result
```

```
C:\Users\terbe\AppData\Local\anaconda3\Lib\site-packages\sklearn\metrics\_classificat
ion.py:1334: UndefinedMetricWarning: Recall is ill-defined and being set to 0.0 due t
o no true samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
C:\Users\terbe\AppData\Local\anaconda3\Lib\site-packages\sklearn\metrics\_classificat
ion.py:1334: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 du
e to no predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
C:\Users\terbe\AppData\Local\anaconda3\Lib\site-packages\sklearn\metrics\_classificat
ion.py:1334: UndefinedMetricWarning: Recall is ill-defined and being set to 0.0 due t
o no true samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
C:\Users\terbe\AppData\Local\anaconda3\Lib\site-packages\sklearn\metrics\_classificat
ion.py:1599: UndefinedMetricWarning: F-score is ill-defined and being set to 0.0 due
to no true nor predicted samples. Use `zero_division` parameter to control this behav
ior.
  _warn_prf(average, "true nor predicted", "F-score is", len(true_sum))
C:\Users\terbe\AppData\Local\anaconda3\Lib\site-packages\sklearn\metrics\_classificat
ion.py:1334: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 du
e to no predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
C:\Users\terbe\AppData\Local\anaconda3\Lib\site-packages\sklearn\metrics\_classificat
ion.py:1334: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 du
e to no predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
C:\Users\terbe\AppData\Local\anaconda3\Lib\site-packages\sklearn\metrics\_classificat
ion.py:1334: UndefinedMetricWarning: Recall is ill-defined and being set to 0.0 due t
o no true samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
C:\Users\terbe\AppData\Local\anaconda3\Lib\site-packages\sklearn\metrics\_classificat
ion.py:1599: UndefinedMetricWarning: F-score is ill-defined and being set to 0.0 due
to no true nor predicted samples. Use `zero_division` parameter to control this behav
ior.
  _warn_prf(average, "true nor predicted", "F-score is", len(true_sum))
```

Out[13]:

	Confusion_Matrix	Accuracy	Precision	Recall	F1 Score	True Positive	False Positive	True Negative	False Negative
0	[[0, 0], [0, 1]]	1.000000	0.000000	0.000000	0.000000	1	0	0	0
1	[[0, 0], [0, 1]]	1.000000	0.000000	0.000000	0.000000	1	0	0	0
2	[[0, 0], [0, 1]]	1.000000	1.000000	1.000000	1.000000	1	0	0	0
3	[[9, 0], [0, 1]]	1.000000	1.000000	1.000000	1.000000	1	0	9	0
4	[[0, 1], [0, 72]]	0.986301	0.986301	1.000000	0.993103	72	1	0	0
5	[[0, 6], [2, 263]]	0.970480	0.977695	0.992453	0.985019	263	6	0	2
6	[[1, 5], [0, 145]]	0.966887	0.966667	1.000000	0.983051	145	5	1	0
7	[[4, 15], [2, 409]]	0.960465	0.964623	0.995134	0.979641	409	15	4	2
8	[[2, 16], [6, 499]]	0.957935	0.968932	0.988119	0.978431	499	16	2	6
9	[[57, 0], [3, 1]]	0.950820	1.000000	0.250000	0.400000	1	0	57	3
10	[[3, 4], [3, 128]]	0.949275	0.969697	0.977099	0.973384	128	4	3	3
11	[[0, 1], [1, 25]]	0.925926	0.961538	0.961538	0.961538	25	1	0	1
12	[[3, 14], [4, 181]]	0.910891	0.928205	0.978378	0.952632	181	14	3	4
13	[[182, 269], [136, 3005]]	0.887249	0.917838	0.956702	0.936867	3005	269	182	136
14	[[1, 16], [2, 135]]	0.883117	0.894040	0.985401	0.937500	135	16	1	2
15	[[21, 41], [14, 364]]	0.875000	0.898765	0.962963	0.929757	364	41	21	14
16	[[7, 83], [10, 632]]	0.872951	0.883916	0.984424	0.931466	632	83	7	10
17	[[57, 86], [35, 762]]	0.871277	0.898585	0.956085	0.926444	762	86	57	35
18	[[197, 29], [1, 0]]	0.867841	0.000000	0.000000	0.000000	0	29	197	1
19	[[1, 20], [7, 174]]	0.866337	0.896907	0.961326	0.928000	174	20	1	7
20	[[8, 22], [8, 179]]	0.861751	0.890547	0.957219	0.922680	179	22	8	8

	Confusion_Matrix	Accuracy	Precision	Recall	F1 Score	True Positive	False Positive	True Negative	False Negative
21	[[140, 259], [158, 2456]]	0.861600	0.904604	0.939556	0.921749	2456	259	140	158
22	[[21, 86], [14, 593]]	0.859944	0.873343	0.976936	0.922240	593	86	21	14
23	[[27, 42], [52, 547]]	0.859281	0.928693	0.913189	0.920875	547	42	27	52
24	[[0, 15], [0, 84]]	0.848485	0.848485	1.000000	0.918033	84	15	0	0
25	[[557, 671], [473, 5525]]	0.841683	0.891704	0.921140	0.906183	5525	671	557	473
26	[[52, 98], [37, 646]]	0.837935	0.868280	0.945827	0.905396	646	98	52	37
27	[[9, 9], [10, 88]]	0.836207	0.907216	0.897959	0.902564	88	9	9	10
28	[[7, 24], [5, 131]]	0.826347	0.845161	0.963235	0.900344	131	24	7	5
29	[[24, 61], [9, 293]]	0.819121	0.827684	0.970199	0.893293	293	61	24	9
30	[[3, 19], [0, 83]]	0.819048	0.813725	1.000000	0.897297	83	19	3	0
31	[[7, 19], [6, 106]]	0.818841	0.848000	0.946429	0.894515	106	19	7	6
32	[[8, 5], [3, 28]]	0.818182	0.848485	0.903226	0.875000	28	5	8	3
33	[[16, 77], [23, 373]]	0.795501	0.828889	0.941919	0.881797	373	77	16	23
34	[[58, 27], [8, 63]]	0.775641	0.700000	0.887324	0.782609	63	27	58	8
35	[[18, 81], [20, 286]]	0.750617	0.779292	0.934641	0.849926	286	81	18	20
36	[[10, 35], [4, 103]]	0.743421	0.746377	0.962617	0.840816	103	35	10	4
37	[[21, 93], [13, 284]]	0.742092	0.753316	0.956229	0.842730	284	93	21	13
38	[[39, 5], [12, 7]]	0.730159	0.583333	0.368421	0.451613	7	5	39	12
39	[[4, 45], [2, 122]]	0.728324	0.730539	0.983871	0.838488	122	45	4	2
40	[[83, 14], [17, 0]]	0.728070	0.000000	0.000000	0.000000	0	14	83	17

	Confusion_Matrix	Accuracy	Precision	Recall	F1 Score	True Positive	False Positive	True Negative	False Negative
41	[[0, 6], [0, 16]]	0.727273	0.727273	1.000000	0.842105	16	6	0	0
42	[[5, 3], [0, 2]]	0.700000	0.400000	1.000000	0.571429	2	3	5	0
43	[[48, 10], [18, 15]]	0.692308	0.600000	0.454545	0.517241	15	10	48	18
44	[[41, 130], [67, 359]]	0.670017	0.734151	0.842723	0.784699	359	130	41	67
45	[[2, 1], [0, 0]]	0.666667	0.000000	0.000000	0.000000	0	1	2	0
46	[[34, 16], [5, 2]]	0.631579	0.111111	0.285714	0.160000	2	16	34	5
47	[[8, 30], [2, 44]]	0.619048	0.594595	0.956522	0.733333	44	30	8	2
48	[[1, 2], [0, 2]]	0.600000	0.500000	1.000000	0.666667	2	2	1	0
49	[[11, 2], [7, 1]]	0.571429	0.333333	0.125000	0.181818	1	2	11	7
50	[[0, 6], [0, 8]]	0.571429	0.571429	1.000000	0.727273	8	6	0	0
51	[[5, 1], [3, 0]]	0.555556	0.000000	0.000000	0.000000	0	1	5	3
52	[[25, 22], [1, 2]]	0.540000	0.083333	0.666667	0.148148	2	22	25	1
53	[[0, 1], [0, 1]]	0.500000	0.500000	1.000000	0.666667	1	1	0	0
54	[[1, 0], [1, 0]]	0.500000	0.000000	0.000000	0.000000	0	0	1	1
55	[[0, 5], [0, 4]]	0.444444	0.444444	1.000000	0.615385	4	5	0	0
56	[[0, 92], [0, 4]]	0.041667	0.041667	1.000000	0.080000	4	92	0	0

In [14]:
```
# Assuming 'Evaluation_Metrics_Fall_2023.xlsx' is the desired file name
result.to_excel('Evaluation_Metrics_StudentLevel_by_Colleger_Fall_2023.xlsx', index=Fa
```

Model Performance Assessment: Predicting Student Continuation in the Fall Semester

Group 1: High Precision and Recall (Ideal)

Graduate Non-Degree Chicago_Pharmacy:

- **Accuracy:** 100%
- **Precision, Recall, F1 Score:** 100%
- *Assessment:* The model excels in accurately predicting students likely to continue their studies in Pharmacy, with no false positives or false negatives. It's an ideal performance.

Graduate Non-Degree Chicago_School of Public Health:

- **Accuracy:** 100%
- **Precision, Recall, F1 Score:** 100%
- *Assessment:* Similar to Pharmacy, the model performs flawlessly in predicting students who continue their studies in the School of Public Health.

Group 2: Balanced Precision and Recall

Professional - Chicago_Dentistry:

- **Accuracy:** 97.0%
- **Balanced Precision (97.8%) and Recall (99.2%)**
- *Assessment:* The model maintains a strong balance between precision and recall, suggesting a reliable identification of students continuing in Dentistry.

Professional - Chicago_College of Medicine Rockford:

- **Accuracy:** 96.7%
- **Balanced Precision (96.7%) and Recall (100%)**
- *Assessment:* The model demonstrates good balance, effectively identifying students continuing in the College of Medicine Rockford.

Group 3: High Accuracy but Potential Issues

Undergrad Non-Degree Chicago_School of Public Health:

- **Perfect Accuracy:** 100%
- **Precision:** 0%
- **Recall:** 0%
- *Assessment:* Although the accuracy is perfect, the model fails to identify any positive instances. This scenario warrants further investigation.

Graduate Non-Degree Chicago_Business Administration:

- **Perfect Accuracy:** 100%
- **Precision:** 0%

- **Recall:** 0%
- *Assessment:* Similar to the previous case, the model shows perfect accuracy but does not capture any positive instances.

Group 4: Trade-off Between Precision and Recall

Graduate Non-Degree Chicago_Nursing:

- **High Accuracy:** 95.1%
- **Precision:** 100%
- **Recall:** 25%
- *Assessment:* The model achieves high precision but at the cost of recall, indicating it's good at correctly identifying positive instances but may miss some.

Overall Summary:

- The model generally performs well, showcasing high accuracy in many categories.
- Ideal scenarios demonstrate perfect precision, recall, and accuracy.
- Balanced precision and recall suggest the model effectively identifies positive instances without excessive false positives or negatives.
- Cases with perfect accuracy but zero precision and recall require scrutiny, as they may indicate issues with the model's predictions.
- Understanding the trade-off between precision and recall is crucial in optimizing the model for specific use cases, such as predicting student continuation in the fall semester.

Undergad

In [15]:
```
undergrad_results_df = coll_student_level_result[coll_student_level_result["Label Name
undergrad_results_df
```

Out[15]:

	Confusion_Matrix	Accuracy	Precision	Recall	F1 Score	True Positive	False Positive	True Negative	False Negative
0	[[0, 0], [0, 1]]	1.000000	0.000000	0.000000	0.000000	1	0	0	0
13	[[182, 269], [136, 3005]]	0.887249	0.917838	0.956702	0.936867	3005	269	182	136
15	[[21, 41], [14, 364]]	0.875000	0.898765	0.962963	0.929757	364	41	21	14
17	[[57, 86], [35, 762]]	0.871277	0.898585	0.956085	0.926444	762	86	57	35
18	[[197, 29], [1, 0]]	0.867841	0.000000	0.000000	0.000000	0	29	197	1
21	[[140, 259], [158, 2456]]	0.861600	0.904604	0.939556	0.921749	2456	259	140	158
25	[[557, 671], [473, 5525]]	0.841683	0.891704	0.921140	0.906183	5525	671	557	473
26	[[52, 98], [37, 646]]	0.837935	0.868280	0.945827	0.905396	646	98	52	37
28	[[7, 24], [5, 131]]	0.826347	0.845161	0.963235	0.900344	131	24	7	5
31	[[7, 19], [6, 106]]	0.818841	0.848000	0.946429	0.894515	106	19	7	6
40	[[83, 14], [17, 0]]	0.728070	0.000000	0.000000	0.000000	0	14	83	17
42	[[5, 3], [0, 2]]	0.700000	0.400000	1.000000	0.571429	2	3	5	0
44	[[41, 130], [67, 359]]	0.670017	0.734151	0.842723	0.784699	359	130	41	67
45	[[2, 1], [0, 0]]	0.666667	0.000000	0.000000	0.000000	0	1	2	0
46	[[34, 16], [5, 2]]	0.631579	0.111111	0.285714	0.160000	2	16	34	5
52	[[25, 22], [1, 2]]	0.540000	0.083333	0.666667	0.148148	2	22	25	1
55	[[0, 5], [0, 4]]	0.444444	0.444444	1.000000	0.615385	4	5	0	0

Model Performance Assessment: Predicting Student Continuation in the Fall Semester

Group 1: High Precision and Recall (Ideal)

Undergrad - Chicago_Engineering:

- **Accuracy:** 88.7%
- **Precision:** 91.8%
- **Recall:** 95.7%
- **F1 Score:** 93.7%
- **True Positive:** 3005
- **False Positive:** 269
- **True Negative:** 182
- **False Negative:** 136
- *Assessment:* The model excels in accurately predicting student continuation in the field of Engineering, achieving high precision and recall.

Undergrad - Chicago_Education:

- **Accuracy:** 87.5%
- **Precision:** 89.9%
- **Recall:** 96.3%
- **F1 Score:** 92.98%
- **True Positive:** 364
- **False Positive:** 41
- **True Negative:** 21
- **False Negative:** 14
- *Assessment:* The model performs well in predicting student continuation in Education, with a good balance between precision and recall.

Undergrad - Chicago_Architecture,Design,& the Arts:

- **Accuracy:** 87.1%
- **Precision:** 89.9%
- **Recall:** 95.6%
- **F1 Score:** 92.64%
- **True Positive:** 762
- **False Positive:** 86
- **True Negative:** 57
- **False Negative:** 35
- *Assessment:* The model effectively predicts student continuation in Architecture, Design, and the Arts, demonstrating balanced precision and recall.

Group 2: Cases with Potential Issues

Undergrad Non-Degree Chicago_VP Undrgrd Affrs:

- **Accuracy:** 86.8%

- **Precision:** 0%
- **Recall:** 0%
- **F1 Score:** 0%
- **True Positive:** 0
- **False Positive:** 29
- **True Negative:** 197
- **False Negative:** 1
- *Assessment:* The model shows perfect accuracy but fails to identify any positive instances, suggesting potential issues with predictions.

Undergrad - Chicago_Business Administration:

- **Accuracy:** 86.2%
- **Precision:** 90.5%
- **Recall:** 93.9%
- **F1 Score:** 92.17%
- **True Positive:** 2456
- **False Positive:** 259
- **True Negative:** 140
- **False Negative:** 158
- *Assessment:* The model performs well but shows a slightly lower recall, indicating it may miss some instances of student continuation in Business Administration.

Undergrad - Chicago_Liberal Arts & Sciences:

- **Accuracy:** 84.2%
- **Precision:** 89.2%
- **Recall:** 92.1%
- **F1 Score:** 90.6%
- **True Positive:** 5525
- **False Positive:** 671
- **True Negative:** 557
- **False Negative:** 473
- *Assessment:* The model performs reasonably well, but the trade-off between precision and recall suggests it may benefit from tuning for specific use cases.

Group 3: Cases with Trade-off Between Precision and Recall

Undergrad - Chicago_Applied Health Sciences:

- **Accuracy:** 83.8%
- **Precision:** 86.8%
- **Recall:** 94.6%

- **F1 Score:** 90.5%
- **True Positive:** 646
- **False Positive:** 98
- **True Negative:** 52
- **False Negative:** 37
- *Assessment:* The model achieves a trade-off between precision and recall, indicating it identifies positive instances well but may miss some.

Undergrad - Chicago_Urban Planning &Public Affairs:

- **Accuracy:** 82.6%
- **Precision:** 84.5%
- **Recall:** 96.3%
- **F1 Score:** 90.03%
- **True Positive:** 131
- **False Positive:** 24
- **True Negative:** 7
- **False Negative:** 5
- *Assessment:* The model demonstrates a good balance, achieving high precision and recall, suggesting reliable predictions.

Undergrad Non-Degree Chicago_Liberal Arts & Sciences:

- **Accuracy:** 72.8%
- **Precision:** 0%
- **Recall:** 0%
- **F1 Score:** 0%
- **True Positive:** 0
- **False Positive:** 14
- **True Negative:** 83
- **False Negative:** 17
- *Assessment:* The model shows perfect accuracy but fails to identify any positive instances, indicating potential issues with predictions.

Undergrad Non-Degree Chicago_Architecture,Design,& the Arts:

- **Accuracy:** 70.0%
- **Precision:** 40%
- **Recall:** 100%
- **F1 Score:** 57.14%
- **True Positive:** 2
- **False Positive:** 3
- **True Negative:** 5
- **False Negative:** 0

- *Assessment:* The model demonstrates a trade-off between precision and recall, suggesting it identifies positive instances well but may have some false positives.

Undergrad - Chicago_Nursing:

- **Accuracy:** 67.0%
- **Precision:** 73.4%
- **Recall:** 84.3%
- **F1 Score:** 78.47%
- **True Positive:** 359
- **False Positive:** 130
- **True Negative:** 41
- **False Negative:** 67
- *Assessment:* The model shows a trade-off between precision and recall, indicating it identifies positive instances well but may miss some.

Undergrad Non-Degree Chicago_Applied Health Sciences:

- **Accuracy:** 66.7%
- **Precision:** 0%
- **Recall:** 0%
- **F1 Score:** 0%
- **True Positive:** 0
- **False Positive:** 1
- **True Negative:** 2
- **False Negative:** 0
- *Assessment:* The model shows perfect accuracy but fails to identify any positive instances, suggesting potential issues with predictions.

Undergrad Non-Degree Chicago_Business Administration:

- **Accuracy:** 63.2%
- **Precision:** 11.1%
- **Recall:** 28.6%
- **F1 Score:** 16%
- **True Positive:** 2
- **False Positive:** 16
- **True Negative:** 34
- **False Negative:** 5
- *Assessment:* The model shows a trade-off between precision and recall, suggesting it identifies positive instances but may have some false positives.

Undergrad Non-Degree Chicago_Engineering:

- **Accuracy:** 54.0%

- **Precision:** 8.3%
- **Recall:** 66.7%
- **F1 Score:** 14.81%
- **True Positive:** 2
- **False Positive:** 22
- **True Negative:** 25
- **False Negative:** 1
- *

Graduate

In [16]:
```
graduate_results_df = coll_student_level_result[coll_student_level_result["Label Name"
graduate_results_df
```

Out[16]:

	Confusion_Matrix	Accuracy	Precision	Recall	F1 Score	True Positive	False Positive	True Negative	False Negative
1	[[0, 0], [0, 1]]	1.000000	0.000000	0.000000	0.000000	1	0	0	0
2	[[0, 0], [0, 1]]	1.000000	1.000000	1.000000	1.000000	1	0	0	0
3	[[9, 0], [0, 1]]	1.000000	1.000000	1.000000	1.000000	1	0	9	0
4	[[0, 1], [0, 72]]	0.986301	0.986301	1.000000	0.993103	72	1	0	0
9	[[57, 0], [3, 1]]	0.950820	1.000000	0.250000	0.400000	1	0	57	3
11	[[0, 1], [1, 25]]	0.925926	0.961538	0.961538	0.961538	25	1	0	1
12	[[3, 14], [4, 181]]	0.910891	0.928205	0.978378	0.952632	181	14	3	4
16	[[7, 83], [10, 632]]	0.872951	0.883916	0.984424	0.931466	632	83	7	10
19	[[1, 20], [7, 174]]	0.866337	0.896907	0.961326	0.928000	174	20	1	7
22	[[21, 86], [14, 593]]	0.859944	0.873343	0.976936	0.922240	593	86	21	14
24	[[0, 15], [0, 84]]	0.848485	0.848485	1.000000	0.918033	84	15	0	0
29	[[24, 61], [9, 293]]	0.819121	0.827684	0.970199	0.893293	293	61	24	9
30	[[3, 19], [0, 83]]	0.819048	0.813725	1.000000	0.897297	83	19	3	0
32	[[8, 5], [3, 28]]	0.818182	0.848485	0.903226	0.875000	28	5	8	3
33	[[16, 77], [23, 373]]	0.795501	0.828889	0.941919	0.881797	373	77	16	23
34	[[58, 27], [8, 63]]	0.775641	0.700000	0.887324	0.782609	63	27	58	8
35	[[18, 81], [20, 286]]	0.750617	0.779292	0.934641	0.849926	286	81	18	20
36	[[10, 35], [4, 103]]	0.743421	0.746377	0.962617	0.840816	103	35	10	4
37	[[21, 93], [13, 284]]	0.742092	0.753316	0.956229	0.842730	284	93	21	13
38	[[39, 5], [12, 7]]	0.730159	0.583333	0.368421	0.451613	7	5	39	12
39	[[4, 45], [2, 122]]	0.728324	0.730539	0.983871	0.838488	122	45	4	2

	Confusion_Matrix	Accuracy	Precision	Recall	F1 Score	True Positive	False Positive	True Negative	False Negative
41	[[0, 6], [0, 16]]	0.727273	0.727273	1.000000	0.842105	16	6	0	0
43	[[48, 10], [18, 15]]	0.692308	0.600000	0.454545	0.517241	15	10	48	18
47	[[8, 30], [2, 44]]	0.619048	0.594595	0.956522	0.733333	44	30	8	2
48	[[1, 2], [0, 2]]	0.600000	0.500000	1.000000	0.666667	2	2	1	0
49	[[11, 2], [7, 1]]	0.571429	0.333333	0.125000	0.181818	1	2	11	7
50	[[0, 6], [0, 8]]	0.571429	0.571429	1.000000	0.727273	8	6	0	0
51	[[5, 1], [3, 0]]	0.555556	0.000000	0.000000	0.000000	0	1	5	3
53	[[0, 1], [0, 1]]	0.500000	0.500000	1.000000	0.666667	1	1	0	0
54	[[1, 0], [1, 0]]	0.500000	0.000000	0.000000	0.000000	0	0	1	1

Model Performance Assessment: Predicting Graduate Student Continuation

Group 1: Ideal Performance

Graduate Non-Degree Chicago_Pharmacy:

- **Accuracy:** 100%
- **Precision:** 100%
- **Recall:** 100%
- **F1 Score:** 100%
- **True Positive:** 1
- **False Positive:** 0
- **True Negative:** 0
- **False Negative:** 0
- *Assessment:* The model achieves perfect performance in predicting student continuation in the Pharmacy program.

Graduate Non-Degree Chicago_School of Public Health:

- **Accuracy:** 100%
- **Precision:** 100%

- **Recall:** 100%
- **F1 Score:** 100%
- **True Positive:** 1
- **False Positive:** 0
- **True Negative:** 9
- **False Negative:** 0
- *Assessment:* The model excels in predicting student continuation in the School of Public Health, achieving perfect precision and recall.

Graduate Non-Degree Chicago_Business Administration:

- **Accuracy:** 100%
- **Precision:** 100%
- **Recall:** 100%
- **F1 Score:** 100%
- **True Positive:** 1
- **False Positive:** 0
- **True Negative:** 0
- **False Negative:** 0
- *Assessment:* The model performs perfectly in predicting student continuation in Business Administration.

Group 2: Strong Performance with Potential Issues

Graduate - Chicago_Coll of Med Office of the Dean:

- **Accuracy:** 98.6%
- **Precision:** 98.6%
- **Recall:** 100%
- **F1 Score:** 99.31%
- **True Positive:** 72
- **False Positive:** 1
- **True Negative:** 0
- **False Negative:** 0
- *Assessment:* The model performs well in predicting student continuation in the College of Medicine Office of the Dean, but the small number of instances might lead to overfitting.

Graduate Non-Degree Chicago_Nursing:

- **Accuracy:** 95.1%
- **Precision:** 100%
- **Recall:** 25%
- **F1 Score:** 40%
- **True Positive:** 1

- **False Positive:** 0
- **True Negative:** 57
- **False Negative:** 3
- *Assessment:* The model shows high precision but low recall, indicating it may miss positive instances in Nursing.

Graduate - Chicago_Graduate College:

- **Accuracy:** 92.6%
- **Precision:** 96.2%
- **Recall:** 96.2%
- **F1 Score:** 96.15%
- **True Positive:** 25
- **False Positive:** 1
- **True Negative:** 0
- **False Negative:** 1
- *Assessment:* The model performs well in predicting student continuation in the Graduate College.

Graduate - Chicago_Nursing:

- **Accuracy:** 91.1%
- **Precision:** 92.8%
- **Recall:** 97.8%
- **F1 Score:** 95.26%
- **True Positive:** 181
- **False Positive:** 14
- **True Negative:** 3
- **False Negative:** 4
- *Assessment:* The model performs well in predicting student continuation in Nursing.

Group 3: Performance with Trade-off Between Precision and Recall

Graduate - Chicago_Liberal Arts & Sciences:

- **Accuracy:** 87.3%
- **Precision:** 88.4%
- **Recall:** 98.4%
- **F1 Score:** 93.15%
- **True Positive:** 632
- **False Positive:** 83
- **True Negative:** 7
- **False Negative:** 10

- *Assessment:* The model demonstrates a good balance between precision and recall in predicting student continuation in Liberal Arts & Sciences.

Graduate - Chicago_Social Work:

- **Accuracy:** 86.6%
- **Precision:** 89.7%
- **Recall:** 96.1%
- **F1 Score:** 92.8%
- **True Positive:** 174
- **False Positive:** 20
- **True Negative:** 1
- **False Negative:** 7
- *Assessment:* The model shows a good balance between precision and recall, indicating reliable predictions in Social Work.

Graduate - Chicago_Engineering:

- **Accuracy:** 85.9%
- **Precision:** 87.3%
- **Recall:** 97.7%
- **F1 Score:** 92.22%
- **True Positive:** 593
- **False Positive:** 86
- **True Negative:** 21
- **False Negative:** 14
- *Assessment:* The model performs well in predicting student continuation in Engineering, with a good balance between precision and recall.

Graduate - Chicago_Architecture,Design,& the Arts:

- **Accuracy:** 84.8%
- **Precision:** 84.8%
- **Recall:** 100%
- **F1 Score:** 91.8%
- **True Positive:** 84
- **False Positive:** 15
- **True Negative:** 0
- **False Negative:** 0
- *Assessment:* The model performs well in predicting student continuation in Architecture, Design, and the Arts, though the small number of instances might lead to overfitting.

Graduate - Chicago_School of Public Health:

- **Accuracy:** 81.9%

- **Precision:** 81.4%
- **Recall:** 100%
- **F1 Score:** 89.73%
- **True Positive:** 293
- **False Positive:** 61
- **True Negative:** 24
- **False Negative:** 9
- *Assessment:* The model demonstrates a good balance between precision and recall, suggesting reliable predictions in the School of Public Health.

Graduate - Chicago_Pharmacy:

- **Accuracy:** 81.9%
- **Precision:** 81.4%
- **Recall:** 100%
- **F1 Score:** 89.73%
- **True Positive:** 83
- **False Positive:** 19
- **True Negative:** 3
- **False Negative:** 0
- *Assessment:* The model shows a good balance between precision and recall, indicating reliable predictions in Pharmacy.

Graduate - Chicago_College of Medicine Rockford:

- **Accuracy:** 81.8%
- **Precision:** 84.8%
- **Recall:** 90.3%
- **F1 Score:** 87.5%
- **True Positive:** 28
- **False Positive:** 5
- **True Negative:** 8
- **False Negative:** 3
- *Assessment:* The model demonstrates a good balance between precision and recall, suggesting reliable predictions in the College of Medicine Rockford.

Graduate - Chicago_Business Administration:

- **Accuracy:** 79.6%
- **Precision:** 82.9%
- **Recall:** 94.2%
- **F1 Score:** 88.18%
- **True Positive:** 373

Professional

In [17]:
```
professional_results_df = coll_student_level_result[coll_student_level_result["Label N
professional_results_df
```

Out[17]:

	Confusion_Matrix	Accuracy	Precision	Recall	F1 Score	True Positive	False Positive	True Negative	False Negative
5	[[0, 6], [2, 263]]	0.970480	0.977695	0.992453	0.985019	263	6	0	2
6	[[1, 5], [0, 145]]	0.966887	0.966667	1.000000	0.983051	145	5	1	0
7	[[4, 15], [2, 409]]	0.960465	0.964623	0.995134	0.979641	409	15	4	2
8	[[2, 16], [6, 499]]	0.957935	0.968932	0.988119	0.978431	499	16	2	6
10	[[3, 4], [3, 128]]	0.949275	0.969697	0.977099	0.973384	128	4	3	3
14	[[1, 16], [2, 135]]	0.883117	0.894040	0.985401	0.937500	135	16	1	2
20	[[8, 22], [8, 179]]	0.861751	0.890547	0.957219	0.922680	179	22	8	8
27	[[9, 9], [10, 88]]	0.836207	0.907216	0.897959	0.902564	88	9	9	10

Model Performance Assessment: Predicting Professional Program Student Continuation

Group 1: Strong Performance

Professional - Chicago_Dentistry:

- **Accuracy:** 97.0%
- **Precision:** 97.8%
- **Recall:** 99.2%
- **F1 Score:** 98.5%
- **True Positive:** 263

- **False Positive:** 6
- **True Negative:** 0
- **False Negative:** 2
- *Assessment:* The model exhibits strong performance in predicting student continuation in Dentistry.

Professional - Chicago_College of Medicine Rockford:

- **Accuracy:** 96.7%
- **Precision:** 96.7%
- **Recall:** 100%
- **F1 Score:** 98.3%
- **True Positive:** 145
- **False Positive:** 5
- **True Negative:** 1
- **False Negative:** 0
- *Assessment:* The model performs well in predicting student continuation in the College of Medicine Rockford.

Professional - Chicago_Pharmacy:

- **Accuracy:** 96.0%
- **Precision:** 96.5%
- **Recall:** 99.5%
- **F1 Score:** 97.96%
- **True Positive:** 409
- **False Positive:** 15
- **True Negative:** 4
- **False Negative:** 2
- *Assessment:* The model exhibits strong performance in predicting student continuation in Pharmacy.

Professional - Chicago_College of Medicine at Chicago:

- **Accuracy:** 95.8%
- **Precision:** 96.9%
- **Recall:** 98.8%
- **F1 Score:** 97.8%
- **True Positive:** 499
- **False Positive:** 16
- **True Negative:** 2
- **False Negative:** 6
- *Assessment:* The model performs well in predicting student continuation in the College of Medicine at Chicago.

Professional - Chicago_Applied Health Sciences:

- **Accuracy:** 94.9%
- **Precision:** 97.0%
- **Recall:** 97.7%
- **F1 Score:** 97.34%
- **True Positive:** 128
- **False Positive:** 4
- **True Negative:** 3
- **False Negative:** 3
- *Assessment:* The model exhibits strong performance in predicting student continuation in Applied Health Sciences.

Group 2: Good Performance with Some Trade-offs

Professional - Chicago_College of Medicine at Chicago:

- **Accuracy:** 88.3%
- **Precision:** 89.4%
- **Recall:** 98.5%
- **F1 Score:** 93.75%
- **True Positive:** 135
- **False Positive:** 16
- **True Negative:** 1
- **False Negative:** 2
- *Assessment:* The model shows good performance but with a small trade-off between precision and recall in predicting student continuation in the College of Medicine at Chicago.

Professional - Chicago_Nursing:

- **Accuracy:** 86.2%
- **Precision:** 89.1%
- **Recall:** 95.7%
- **F1 Score:** 92.27%
- **True Positive:** 179
- **False Positive:** 22
- **True Negative:** 8
- **False Negative:** 8
- *Assessment:* The model demonstrates good performance but with a trade-off between precision and recall in predicting student continuation in Nursing.

Group 3: Performance with Trade-off Between Precision and Recall

Professional - Chicago_School of Public Health:

- **Accuracy:** 83.6%
- **Precision:** 90.7%
- **Recall:** 89.8%
- **F1 Score:** 90.26%
- **True Positive:** 88
- **False Positive:** 9
- **True Negative:** 9
- **False Negative:** 10
- *Assessment:* The model exhibits a trade-off between precision and recall in predicting student continuation in the School of Public Health.

Law

In [18]:
```
law_results_df = coll_student_level_result[coll_student_level_result["Label Name"].str
law_results_df
```

Out[18]:

	Confusion_Matrix	Accuracy	Precision	Recall	F1 Score	True Positive	False Positive	True Negative	False Negative
23	[[27, 42], [52, 547]]	0.859281	0.928693	0.913189	0.920875	547	42	27	52

Model Performance Assessment: Predicting Law Program Student Continuation

Group 1: Strong Performance

Law - Chicago_UIC School of Law:

- **Accuracy:** 85.9%
- **Precision:** 92.9%
- **Recall:** 91.3%
- **F1 Score:** 92.1%
- **True Positive:** 547
- **False Positive:** 42
- **True Negative:** 27

- **False Negative:** 52
- *Assessment:* The model exhibits strong performance in predicting student continuation in the UIC School of Law.

Non- Credit

In [19]:
```
non_credit_results_df = coll_student_level_result[coll_student_level_result["Label Nam
non_credit_results_df
```

Out[19]:

	Confusion_Matrix	Accuracy	Precision	Recall	F1 Score	True Positive	False Positive	True Negative	False Negative	Labe
56	[[0, 92], [0, 4]]	0.041667	0.041667	1.0	0.08	4	92	0	0	No Chi fo Enga

Model Performance Assessment: Predicting Non-Credit Program Student Continuation

Group 2: Poor Performance

Non-Credit - Chicago_VP for Global Engagement:

- **Accuracy:** 4.2%
- **Precision:** 4.2%
- **Recall:** 100%
- **F1 Score:** 8.0%
- **True Positive:** 4
- **False Positive:** 92
- **True Negative:** 0
- **False Negative:** 0
- *Assessment:* The model shows poor performance in predicting student continuation in the Non-Credit program under the VP for Global Engagement.

In [ ]: